Light

GCSE Computer Science - Memory and Storage (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Memory And Storage

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Why does a computer need RAM in addition to secondary storage?

- ○ **A.** The CPU can only directly access data in RAM, not from secondary storage
- ○ **B.** RAM provides permanent backup of all files
- ○ **C.** RAM is cheaper than secondary storage
- ○ **D.** RAM stores the BIOS instructions

Q2 (1 mark)

What would happen if a computer had no ROM?

- ○ **A.** The computer would not know how to start up when powered on
- ○ **B.** The computer could not connect to the internet
- ○ **C.** The computer would run more slowly
- ○ **D.** The computer would lose all saved files

Q3 (1 mark)

Why is cache memory faster than RAM?

- ○ **A.** Cache uses magnetic storage technology
- ○ **B.** Cache is connected directly to the internet
- ○ **C.** Cache has larger storage capacity
- ○ **D.** Cache is positioned closer to the CPU and uses faster memory technology

Q4 (1 mark)

What type of data do CPU registers typically hold?

- ○ **A.** The computer's entire operating system
- ○ **B.** All of the user's saved documents
- ○ **C.** The data and instructions the CPU is currently processing
- ○ **D.** Backup copies of the hard drive

Q5 (1 mark)

Which of the following are all types of secondary storage?

- ○ **A.** Hard disk drive, solid state drive, and optical disc
- ○ **B.** RAM, cache, and registers
- ○ **C.** ROM, BIOS, and firmware
- ○ **D.** CPU, GPU, and motherboard

Q6 (1 mark)

Why do SSDs consume less power than HDDs during operation?

- ○ **A.** SSDs have larger storage capacities
- ○ **B.** SSDs are always connected to external power supplies
- ○ **C.** SSDs automatically turn off when not in use
- ○ **D.** SSDs have no motors or moving parts that require power to operate

Q7 (1 mark)

Why do HDDs make noise during operation while SSDs are silent?

- ○ **A.** HDDs produce sounds from their spinning platters and moving heads, while SSDs have no moving parts
- ○ **B.** HDDs have built-in speakers for alerts

- ○ **C.** SSDs are wrapped in soundproofing material
- ○ **D.** HDDs are always larger and heavier

Q8 (1 mark)

Where are files physically stored when using cloud storage?

- ○ **A.** In the atmosphere among actual clouds
- ○ **B.** On the user's local hard drive only
- ○ **C.** In the computer's RAM
- ○ **D.** On remote servers in data centres operated by the cloud provider

Q9 (1 mark)

What is the practical consequence of RAM being volatile?

- ○ **A.** RAM operates at dangerously high temperatures
- ○ **B.** Work must be saved to secondary storage or it will be lost when power is removed
- ○ **C.** RAM cannot store numerical data
- ○ **D.** RAM must be replaced every month

Q10 (1 mark)

Which statement correctly identifies non-volatile storage only?

- ○ **A.** ROM, SSD, and HDD are non-volatile
- ○ **B.** Registers and ROM are non-volatile
- ○ **C.** RAM and cache are non-volatile
- ○ **D.** Cache and SSD are non-volatile

Q11 (1 mark)

Why is there a hierarchy of different memory types rather than using the fastest memory for everything?

- ○ **A.** Slower memory produces better image quality
- ○ **B.** Faster memory is significantly more expensive, making large quantities unaffordable
- ○ **C.** Computers can only support one type of fast memory
- ○ **D.** Slower memory is more colourful

Q12 (1 mark)

Where is virtual memory physically located?

- ○ **A.** Inside the CPU
- ○ **B.** On the graphics card
- ○ **C.** In the cloud
- ○ **D.** On the hard drive or SSD as a designated swap file or partition

Q13 (2 marks)

A student claims that ROM is unnecessary because RAM can store the same data. Why is this incorrect?

- ○ **A.** ROM is faster than RAM
- ○ **B.** ROM has larger storage capacity than RAM
- ○ **C.** RAM is empty when power is first applied, so the computer needs non-volatile ROM to provide initial startup instructions
- ○ **D.** ROM is cheaper than RAM

Q14 (2 marks)

When the CPU needs data, it first checks the cache. What happens if the required data is found there (a 'cache hit')?

- ○ **A.** The data is retrieved almost instantly, avoiding the slower process of accessing main memory
- ○ **B.** The CPU must wait for the data to be verified by the hard drive
- ○ **C.** The data is deleted from the cache to make room for new data
- ○ **D.** The data is copied to ROM for permanent storage

Q15 (2 marks)

A computer with an HDD takes 90 seconds to fully boot into the operating system. After upgrading to an SSD, boot time drops to 15 seconds. What explains this dramatic improvement?

- ○ **A.** SSDs have more storage capacity

- ○ **B.** SSDs eliminate the mechanical delays of spinning platters and moving heads, providing much faster data transfer rates for loading the operating system
- ○ **C.** SSDs compress the operating system to make it smaller
- ○ **D.** SSDs bypass the need to load an operating system entirely

Q16 (2 marks)

Why do HDDs benefit from periodic defragmentation while SSDs do not need it?

- ○ **A.** SSDs cannot store fragmented files
- ○ **B.** SSDs automatically defragment themselves every hour
- ○ **C.** Defragmentation only works on magnetic storage
- ○ **D.** HDDs are slower when reading fragmented files because the head must move to multiple locations, but SSD access time is constant regardless of data location

Q17 (2 marks)

Why is cloud storage often recommended as part of a backup strategy?

- ○ **A.** Cloud storage is faster than local storage
- ○ **B.** Cloud files are stored in a separate physical location, protecting against local disasters like fire, theft, or hardware failure
- ○ **C.** Cloud storage makes files invisible to hackers
- ○ **D.** Cloud storage compresses files to use less space

Q18 (2 marks)

A user has 500GB of photos they want to upload to cloud storage. Their internet connection has an upload speed of 10 Mbps. What practical limitation will they encounter?

- ○ **A.** Cloud storage cannot accept photo files
- ○ **B.** The upload will take many days because internet bandwidth limits how quickly large amounts of data can be transferred to remote servers
- ○ **C.** The photos will lose quality when uploaded
- ○ **D.** Only 100GB can be uploaded to any cloud service

Q19 (2 marks)

A CPU register can hold 8 bytes while RAM can hold 16GB. Despite this huge size difference, why are registers essential?

- ○ **A.** Registers store data permanently while RAM is temporary
- ○ **B.** Registers are much cheaper than RAM
- ○ **C.** Registers can store more complex data types
- ○ **D.** Registers provide instant access for data the CPU is actively processing, as accessing RAM would cause delays that slow down every operation

Q20 (2 marks)

When the CPU needs to execute a program stored on the hard drive, what must happen first?

- ○ **A.** The CPU reads directly from the hard drive

- ○ **B.** The program must be copied from the hard drive into RAM before the CPU can execute it
- ○ **C.** The hard drive converts the program into electrical signals
- ○ **D.** The program is uploaded to cloud storage for processing

Q21 (2 marks)

SSD flash memory cells have a limited number of write cycles before wearing out. Why is this usually not a concern for typical users?

- ○ **A.** Modern SSDs have write endurance ratings that exceed the amount most users will write over the drive's practical lifespan
- ○ **B.** SSDs automatically repair worn cells
- ○ **C.** Users never write data to SSDs
- ○ **D.** Write limitations only apply to older technology

Q22 (2 marks)

A student is writing an essay on their computer. They have typed 1000 words but have not yet clicked 'Save'. What is the current location of their work?

- ○ **A.** Permanently stored on the hard drive
- ○ **B.** Stored only in volatile RAM and will be lost if power is interrupted
- ○ **C.** Backed up automatically to cloud storage
- ○ **D.** Stored in ROM for safety

Q23 (2 marks)

When a computer relies heavily on virtual memory, users often notice the hard drive activity light staying constantly lit and system responsiveness becoming very poor. What causes this?

- ○ **A.** The hard drive is overheating
- ○ **B.** The computer is downloading files from the internet
- ○ **C.** The system is continuously swapping data between RAM and storage because RAM is insufficient, and the slow storage access creates delays
- ○ **D.** The activity light is malfunctioning

Q24 (3 marks)

A hospital needs to store patient records that must be: accessible quickly in emergencies, protected against loss from hardware failure, compliant with data protection regulations requiring records stay within the country, and accessible from multiple departments. Analyse which storage solution best meets these requirements.

- ○ **A.** USB flash drives distributed to each department for portability
- ○ **B.** A single large HDD for cost efficiency
- ○ **C.** A local SSD-based server with RAID redundancy for speed and protection against drive failure, combined with secure local backup systems to maintain data within national jurisdiction while enabling multi-department network access
- ○ **D.** Public cloud storage only, as it provides the easiest access

Q25 (3 marks)

Explain how the memory hierarchy (registers, cache, RAM, secondary storage) works together during program execution to balance speed with capacity and cost.

- ○ **A.** Each memory type operates independently with no interaction between levels
- ○ **B.** All data is stored in registers because they are fastest; other memory types are unused during execution
- ○ **C.** Secondary storage sends data directly to registers, bypassing other memory levels for efficiency
- ○ **D.** The hierarchy creates a cascade where the CPU works directly with small, fast registers; cache stores recently used data from RAM for quick access; RAM holds active programs copied from storage; and secondary storage maintains permanent files, with data moving between levels as needed

Q26 (3 marks)

A music producer creates and edits audio files that are typically 2-5GB each. They need to collaborate with musicians remotely, access projects from both their studio and home computers, and maintain backups of all work. Analyse the optimal storage approach.

- ○ **A.** Work exclusively from cloud storage to maximise accessibility
- ○ **B.** Use only local HDD storage to maximise capacity
- ○ **C.** Use a local SSD for active project work providing the fast access needed for audio editing, synchronise completed projects and backups to cloud storage for remote access and protection, and maintain a local backup drive for quick recovery
- ○ **D.** Store everything on USB flash drives for maximum portability

Q27 (3 marks)

A computer with 8GB RAM is upgraded to 32GB. Analyse how this affects performance for: (1) web browsing with many tabs, (2) loading the operating system at startup, and (3) the speed of mathematical calculations in a spreadsheet.

- ○ **A.** Heavy web browsing improves significantly as more tabs can stay in RAM without swapping; boot time may improve slightly if the OS no longer competes with cached data for space; calculation speed is unchanged as computations occur in the CPU, not RAM
- ○ **B.** None of the three activities are affected because RAM capacity has no impact on performance
- ○ **C.** All three activities become dramatically faster because more RAM increases all aspects of computer speed
- ○ **D.** Only boot time improves; web browsing and calculations are unaffected by RAM

Q28 (3 marks)

Some computers use a configuration with a small SSD and a large HDD together. Explain the logic behind this combination and how it addresses different storage needs.

- ○ **A.** This configuration is inefficient; computers should use only one storage type
- ○ **B.** The HDD stores the operating system while the SSD stores games, optimising for different access patterns
- ○ **C.** The small SSD stores the operating system and frequently-used applications for fast boot and loading times, while the large HDD provides affordable bulk storage for documents, media, and less-frequently-accessed files
- ○ **D.** The SSD and HDD must be exactly the same size to function together

Q29 (3 marks)

Modern CPUs have multiple cache levels (L1, L2, L3) rather than a single large cache. Explain why this multi-level design provides better performance than one large cache would.

- ○ **A.** Multiple levels are purely marketing; they provide no performance advantage
- ○ **B.** Each level stores different file types: L1 for documents, L2 for images, L3 for videos
- ○ **C.** L3 cache is fastest while L1 is slowest, creating an inverted hierarchy for efficiency
- ○ **D.** L1 cache is extremely fast but tiny, providing instant access for the most critical data; L2 is larger but slightly slower; L3 is largest but slowest - this hierarchy ensures frequently-needed data has fastest possible access while still providing good hit rates for broader data sets

Q30 (3 marks)

A smart thermostat is an embedded system that must: start instantly when power is applied, operate reliably for years without maintenance, and store user temperature preferences even during power outages. Analyse why it uses ROM and flash memory rather than traditional RAM and HDD storage.

- ○ **A.** RAM and HDD are too expensive for embedded devices
- ○ **B.** ROM provides the program code that executes immediately at power-on without loading time; flash memory stores user settings persistently without power; this combination eliminates the boot process needed by RAM-based systems and the mechanical failure risk of HDDs
- ○ **C.** Embedded systems cannot use any form of electronic memory

○ **D.** The thermostat requires a keyboard and monitor that RAM provides

Total: **55** marks • Time: **30 min**

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